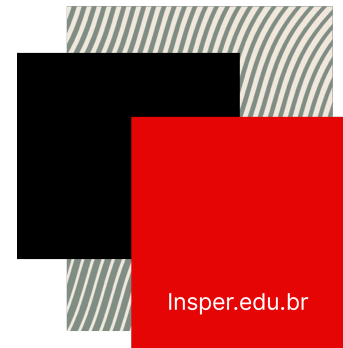
Economics of Organizations and Incentives

MPE – 4th term, 2026

Lecture 2: Static games of complete information II



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Recap: normal-form games and dominance

- A normal-form game is $G = (I,\ \{S_i\}_{i \in I},\ \{u_i\}_{i \in I})$: players, strategy sets and payoff functions.

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- **Lesson 1:** a rational player never plays a strictly dominated strategy.
- **IESDS:** if rationality is common knowledge, strictly dominated strategies can be eliminated iteratively.

Recap: where to open the pharmacy?

- Two chains choosing among 7 blocks: blocks 1 and 7 are strictly dominated; after three rounds of IESDS, only $(4, 4)$ survives.

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Recap: where to open the pharmacy?

- Two chains choosing among 7 blocks: blocks 1 and 7 are strictly dominated; after three rounds of IESDS, only $(4, 4)$ survives.
- Each round requires one more level of "I know that you know that I am rational".
- The prediction was sharp because prices were fixed. If the chains also chose prices, they would want to move apart (we will see this in the problem set).

Back to Lecture 1: the prisoner's dilemma

Two suspects are questioned separately. Each can confess (*fink*, F) in exchange for a reduced sentence, or stay silent (*mum*, M).

		Player 2	
		M	F
Player 1	M	$-2,\ -2$	$-5,\ -1$
	F	$-1,\ -5$	$-4,\ -4$

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For each player, F strictly dominates M: confessing is better whatever the other suspect does.

Dominant-strategy equilibrium

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In the prisoner's dilemma, (F, F) is the strict dominant-strategy equilibrium.

Dominance and social dilemmas

Definition (Pareto optimality)

A strategy profile $s \in S$ **Pareto dominates** $s' \in S$ if $u_i(s) \geq u_i(s')$ for all $i \in I$ and $u_i(s) > u_i(s')$ for at least one $i \in I$. A strategy profile is **Pareto optimal** if no other strategy profile Pareto dominates it.

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Lesson 2: individually rational choices can lead to socially undesirable outcomes.

Epistemic interpretation of IESDS

- S_i^1: strategies compatible with **rationality** (R).
- S_i^2: compatible with rationality and with the belief that others are rational.
- S_i^k: compatible with R and $k-1$ levels of belief in rationality.
- S_i^∞: compatible with **common belief in rationality**.

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- S_i^∞: compatible with **common belief in rationality**.

But what if no strategy is dominated?

Best responses and rationalizability

Example 1: battle of the sexes

Alex and Chris must choose where to meet without communicating. Both prefer to be together; Alex prefers the opera (O) and Chris prefers football (F). Being apart is the worst outcome for both.

		Chris: O	Chris: F
Alex	O	$2, 1$	$0, 0$
	F	$0, 0$	$1, 2$

Is there a **dominant** strategy? Are there **dominated** strategies? What does IESDS predict?

Best response

Definition (best response)

$s_i \in S_i$ is a **best response** of player i to $s_{-i} \in S_{-i}$ if $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i})$ for all $s'_i \in S_i$.

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Definition (best-response correspondence)

$BR_i(s_{-i}) \subseteq S_i$ is the set of all best responses of player i to s_{-i}.

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Definition (best-response correspondence)

$BR_i(s_{-i}) \subseteq S_i$ is the set of all best responses of player i to s_{-i}.

Definition (never a best response)

s_i is **never a best response** (NBR) if there is no belief s_{-i} for which $s_i \in BR_i(s_{-i})$.

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- A rational player only chooses strategies that are a best response to some belief.
- Using common knowledge of rationality, we can iterate: eliminate every strategy that is never a best response…
- …which yields a smaller game, where we eliminate again, and again.

The strategies that survive this process are the **rationalizable strategies**.

Rationalizability

Proposition

In a finite normal-form game, if s^* is a strict dominant-strategy equilibrium, or if s^* uniquely survives IESDS, then $s_i^* \in BR_i(s_{-i}^*)$ for every $i \in I$.

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Lesson 3: a rational player never plays a strategy that is never a best response.

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Lesson 3: a rational player never plays a strategy that is never a best response.

Lesson 4: a rational player who believes that the opponents play s_{-i} chooses a best response to s_{-i}.

Nash equilibrium

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Definition

A **Nash equilibrium** of G is a strategy profile $s^* = (s_1^*, \ldots, s_n^*)$ such that, for every $i \in I$ and every $s_i \in S_i$,

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In words: each player's strategy is a best response to the others' strategies, so no player gains from a *unilateral* deviation.

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In words: each player's strategy is a best response to the others' strategies, so no player gains from a *unilateral* deviation.

In the battle of the sexes: (O, O) and (F, F) are Nash equilibria.

Nash equilibrium vs. iterated elimination

- IESDS and rationalizability are *epistemic*: they follow from common knowledge of rationality.
- Nash equilibrium is a notion of *strategic stability*: beliefs are correct.

Nash equilibrium vs. iterated elimination

- IESDS and rationalizability are *epistemic*: they follow from common knowledge of rationality.
- Nash equilibrium is a notion of *strategic stability*: beliefs are correct.
- Interpretations of a Nash equilibrium:
  - self-fulfilling expectations;
  - self-enforcing agreements;
  - social norms or conventions.

Nash equilibrium and IESDS

Proposition 1

If s^* is a Nash equilibrium, then s^* survives IESDS.

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Proof. Suppose not, and consider the *first* round k in which some s_i^* is eliminated.

- Then s_i^* is strictly dominated by some $s_i' \in S_i^{k-1}$ against every $s_{-i} \in S_{-i}^{k-1}$.
- No component of s^* was eliminated before round k, so $s_{-i}^* \in S_{-i}^{k-1}$. Hence $u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$.
- So s_i^* is not a best response to s_{-i}^*: a contradiction with s^* being a Nash equilibrium. □

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Note: a Nash equilibrium may involve *weakly* dominated strategies; such equilibria are often considered less plausible.

Nash equilibrium and IESDS

Proposition 2

In a finite game, if s^* is the unique profile that survives IESDS, then s^* is a Nash equilibrium (and, by Proposition 1, the unique one).

Nash equilibrium and IESDS

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In a finite game, if s^* is the unique profile that survives IESDS, then s^* is a Nash equilibrium (and, by Proposition 1, the unique one).

Proof. Suppose some player i has s_i with $u_i(s_i, s^*_{-i}) > u_i(s^*_i, s^*_{-i})$.

- $s_i \neq s^*_i$ was eliminated in some round, by some s'_i that strictly dominates it against all surviving s_{-i}, including s^*_{-i}: $u_i(s'_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$.
- If s'_i is eliminated later, repeat the argument. Since the game is finite, the chain ends at a strategy that is never eliminated, that is, at s^*_i.
- Payoffs against s^*_{-i} strictly increase along the chain, so $u_i(s^*_i, s^*_{-i}) > u_i(s_i, s^*_{-i})$: a contradiction. □

Examples: coordination and dilemmas

Which side of the road?

	L	R
L	$1,\ 1$	$0,\ 0$
R	$0,\ 0$	$1,\ 1$

Nash equilibria: (L, L) and (R, R).
IESDS eliminates nothing.

Prisoner's dilemma

	C	D
C	$2,\ 2$	$-1,\ 3$
D	$3,\ -1$	$0,\ 0$

Unique Nash equilibrium: (D, D), although (C, C) Pareto dominates it.

Oligopoly: Cournot

Example 2: Cournot duopoly

- Two firms produce a homogeneous good and choose outputs $q_i \geq 0$ simultaneously.
- Inverse demand: $P(Q) = 100 - Q$, with $Q = q_1 + q_2$.
- Cost: $C(q_i) = 10\,q_i$ for both firms.

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Find:

(a) the normal form of the game and each firm's payoff function;

(b) the dominant and dominated strategies, if any;

(c) each firm's best-response function; plot both in the same graph;

(d) the Nash equilibrium. How does it compare with the monopoly output?

Cournot duopoly: the model

- Players: $I = \{1, 2\}$. Strategies: quantities $S_i = \{q_i : q_i \geq 0\}$.
- Homogeneous good: consumers only care about the price, which is the same for both firms and is given by the inverse demand

$$P(q_1 + q_2) = a - b\,(q_1 + q_2).$$

- Each firm maximizes its own profit. With constant marginal cost c,

$$\Pi_i(q_i, q_j) = P(q_i + q_j)\,q_i - c\,q_i = \big(a - c - b(q_i + q_j)\big)\,q_i.$$

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Example: $P(Q) = 100 - Q$ and $c = 10$ give $\Pi_i = (100 - q_i - q_j)\,q_i - 10\,q_i$.

Insper

Two benchmarks

Monopoly (a single firm):

$$RM(q^m) = c \;\Rightarrow\; q^m = \frac{a-c}{2b}.$$

Example: $q^m = 45$.

Perfect competition (many identical firms):

$$P(q^{cp}) = c \;\Rightarrow\; q^{cp} = \frac{a-c}{b} = 2\,q^m.$$

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Where does the duopoly fall between these two benchmarks?

Cournot duopoly: best response

Given q_j, firm i solves $\max_{q_i \geq 0} \; (a - c - b(q_i + q_j))\, q_i$. First-order condition:

$$\frac{\partial \Pi_i}{\partial q_i} = a - c - 2b\, q_i - b\, q_j = 0.$$

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Best response (with the corner solution):

$$q_i^*(q_j) = \begin{cases} q^m - \frac{1}{2}\, q_j & \text{if } 0 \leq q_j < q^{cp}, \\ 0 & \text{if } q_j \geq q^{cp}. \end{cases}$$

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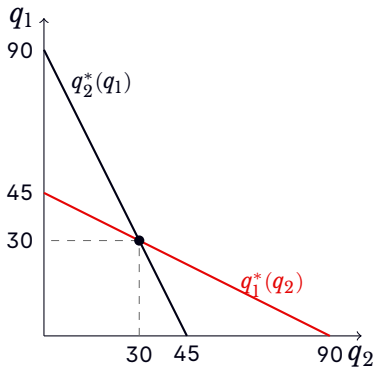
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Example: $100 - 2q_i - q_j - 10 = 0 \ \Rightarrow \ q_i^*(q_j) = \dfrac{90 - q_j}{2}$.

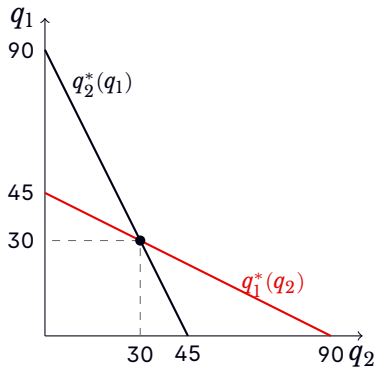
Cournot duopoly: Nash equilibrium



A Nash equilibrium is a pair of mutual best responses:

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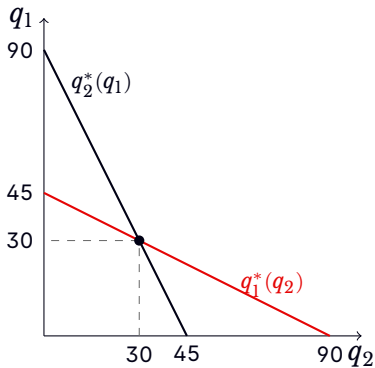


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IESDS gives the same answer: eliminating outputs that are never optimal yields the sequence $\frac{1}{2}q^m, \frac{3}{4}q^m, \frac{5}{8}q^m, \frac{11}{16}q^m, \cdots \to \frac{2}{3}q^m$.

Cournot duopoly: cartel

- A **cartel**: competing firms coordinate to maximize joint profit, as a single monopolist.
- With identical costs, each firm produces $q^m/2$ (in the example, 22.5).

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		Firm 2: Cartel	Firm 2: Deviate
Firm 1	Cartel	$1{,}012,\ 1{,}012$	$759,\ 1{,}139$
	Deviate	$1{,}139,\ 759$	$900,\ 900$

Deviate = best response to the cartel output, $q_i^*(22.5) = 33.75$. Which game has the same structure?

Cournot oligopoly with n firms

Price: $P(q_1 + \cdots + q_n) = a - b\,(q_1 + \cdots + q_n)$; payoff of firm i: $q_i \left[P\left(\sum_j q_j\right) - c\right]$.

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Best response: $q_i^*(q_{-i}) = q^m - \frac{1}{2}\sum_{j \neq i} q_j$ (if positive). In equilibrium, for every i,

$$q_i^* = q^m - \tfrac{1}{2}\sum_{j \neq i} q_j^* \;\Rightarrow\; q_i^* = 2q^m - \sum_{j \in I} q_j^* = 2q^m - Q^*.$$

Cournot oligopoly with n firms

From the previous slide, $q_i^* = 2q^m - Q^*$ for every i.

Summing over $i \in I$: $Q^* = 2n\,q^m - n\,Q^* \;\Rightarrow\; Q^* = \dfrac{2n}{n+1}\,q^m$. Substituting back,

$$q_1^* = \cdots = q_n^* = \frac{2\,q^m}{n+1}.$$

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As $n \to \infty$, $Q^* \to 2q^m = q^{cp}$: with many identical firms, the market approaches perfect competition.

Oligopoly: Bertrand

Bertrand duopoly: the model

In many markets, firms compete in *prices* rather than quantities.

- Two firms produce a homogeneous good and choose prices $p_i \geq 0$ simultaneously; marginal cost $c \geq 0$ for both.
- Consumers always buy from the cheapest firm. Market demand at the lowest price p: $Q(p) = \dfrac{a - p}{b}$ if $p < a$, and 0 otherwise.

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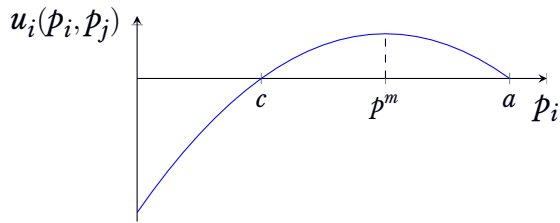
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Payoff of firm i:

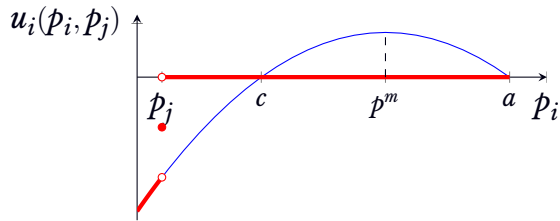
$$u_i(p_i, p_j) = \begin{cases} (p_i - c)\, Q(p_i) & \text{if } p_i < p_j, \\ \frac{1}{2}(p_i - c)\, Q(p_i) & \text{if } p_i = p_j, \\ 0 & \text{if } p_i > p_j. \end{cases}$$

Bertrand duopoly: the monopoly price

The monopoly price maximizes $(p - c)\,Q(p) \implies p^m = \dfrac{a+c}{2}$.



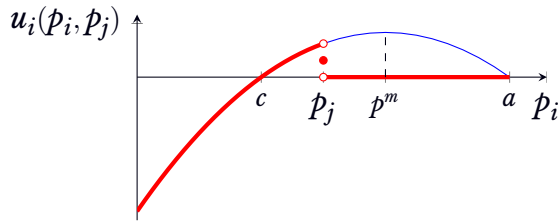
Bertrand duopoly: payoff of i given p_j



Case $p_j \leq c$

Any $p_i < p_j$ yields a loss, and any $p_i > p_j$ yields zero profit, simply because i does not sell.

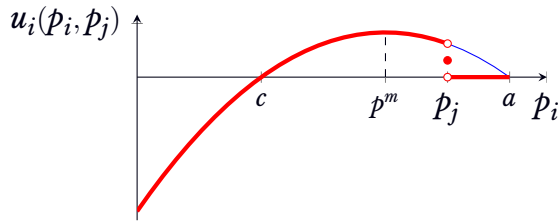
Bertrand duopoly: payoff of i given p_j



Case $c < p_j \leq p^m$

Firm i wants to choose $p_i < p_j$, as close as possible to p_j. Existence problem: for any $p_i < p_j$ there is a higher price, still below p_j, that raises the firm's profit.

Bertrand duopoly: payoff of i given p_j



Case $p^m < p_j < a$

There is a unique best response: the monopoly price. Firm i takes the whole market with any price below p_j, and when it can do so, the best choice is p^m.

Bertrand duopoly: best response

Best response of firm i to p_j:

$$p_i^*(p_j) = \begin{cases} \{p_i : p_i > p_j\} & \text{if } p_j < c, \\ \{p_i : p_i \geq p_j\} & \text{if } p_j = c, \\ \emptyset & \text{if } c < p_j \leq p^m, \\ \{p^m\} & \text{if } p^m < p_j. \end{cases}$$

Bertrand duopoly: Nash equilibrium

$(p_1^*, p_2^*) = (c, c)$ is a Nash equilibrium.

- Suppose that $p_2 = c$.

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- In particular, c is a best response. The same argument holds for firm 2.

Bertrand duopoly: Nash equilibrium

- Note that $p_i^* = c$ is a *weakly dominated* strategy. (**Why?**)

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- We found one Nash equilibrium. Are there others? (**No: we show it next.**)

Bertrand duopoly: (c, c) is the unique equilibrium

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- An analogous argument shows that we cannot have $p_1^* < p_2^*$.

Bertrand duopoly: (c, c) is the unique equilibrium

Step 2. If (p_1^*, p_2^*) is a Nash equilibrium, then $p_1^* = p_2^* = c$.

- If $p_1^* = p_2^* > c$, firm 1 could charge $p_1' \in (c, p_2^*)$ and serve the whole market instead of sharing it.

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According to the model, a duopoly in Nash equilibrium works like a competitive market.

Bertrand paradox

- A duopoly behaves as if it were in perfect competition: price equals marginal cost and profits are zero.

Bertrand paradox

- A duopoly behaves as if it were in perfect competition: price equals marginal cost and profits are zero.
- Assumptions behind the result: (i) homogeneous products; (ii) simultaneous choices; (iii) unlimited production capacity.

Cournot or Bertrand?

Which model should we use? It depends on what is harder to adjust.

- **Cournot:** quantity (capacity) is harder to adjust than price. The strategic decision is how much to produce. Examples: cars, cement, steel, commodities.
- **Bertrand:** price is harder to adjust than quantity, or quantity can be expanded at almost no cost. The strategic decision is the price. Example: software and other digital goods.

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In both models, the Nash equilibrium predicts *more* output than a cartel, but the Bertrand prediction is much more extreme.

Mixed strategies

Mixed strategies

Definition

A **mixed strategy** of player i is a probability distribution over S_i: $\sigma_i : S_i \to [0,1]$ such that $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$. We denote the set of mixed strategies by $\Sigma_i := \Delta(S_i)$, and a mixed-strategy profile by $\sigma = (\sigma_1, \ldots, \sigma_n) \in \Sigma := \Sigma_1 \times \cdots \times \Sigma_n$.

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- A player randomizes to avoid being predictable: to surprise, or to avoid being surprised.
- Every pure strategy is a (degenerate) mixed strategy.
- With mixed strategies, players evaluate *expected* payoffs.

Mixed strategies: expected payoffs

Payoff of i given a profile σ:

$$u_i(\sigma) = \sum_{s \in S} \left[\sigma_1(s_1) \cdots \sigma_n(s_n)\right] u_i(s).$$

Mixed strategies: expected payoffs

Payoff of i given a profile σ:

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Payoff of i from the pure strategy s_i, given σ_{-i}:

$$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \left[\prod_{j \neq i} \sigma_j(s_j)\right] u_i(s_i, s_{-i}).$$

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Mixed-strategy Nash equilibrium

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A **Nash equilibrium** is a profile $\sigma^* \in \Sigma$ such that $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)$ for every $\sigma_i \in \Sigma_i$ and every $i \in I$.

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Principle of indifference

$\sigma^* \in \Sigma$ is a Nash equilibrium if and only if, for every $i \in I$,

1. $u_i(\sigma^*) = u_i(s_i, \sigma_{-i}^*)$ for every $s_i \in \operatorname{supp} \sigma_i^*$;
2. $u_i(\sigma^*) \geq u_i(s_i, \sigma_{-i}^*)$ for every $s_i \in S_i \setminus \operatorname{supp} \sigma_i^*$,

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where $\operatorname{supp} \sigma_i := \{s_i \in S_i : \sigma_i(s_i) > 0\}$ is the support of σ_i.

In words: every strategy played with positive probability yields the same expected payoff, and no strategy outside the support yields more.

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Principle of indifference: proof

[Indifference $\Rightarrow$ Nash] If σ^* satisfies (1) and (2) for every i, then for every $\sigma_i \in \Sigma_i$,

$$u_i(\sigma_i, \sigma^*_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(s_i, \sigma^*_{-i}) \;\leq\; \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(\sigma^*_i, \sigma^*_{-i}) = u_i(\sigma^*_i, \sigma^*_{-i}).$$

Principle of indifference: proof

[Nash $\Rightarrow$ indifference] Suppose σ^* is a Nash equilibrium but (1) fails: there are $s_j', s_j'' \in \operatorname{supp} \sigma_j^*$ with $u_j(s_j'', \sigma_{-j}^*) > u_j(s_j', \sigma_{-j}^*)$. Consider

$$\hat{\sigma}_j(s_j) = \begin{cases} \sigma_j^*(s_j) & \text{if } s_j \notin \{s_j', s_j''\}, \\ 0 & \text{if } s_j = s_j', \\ \sigma_j^*(s_j') + \sigma_j^*(s_j'') & \text{if } s_j = s_j''. \end{cases}$$

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Moving the probability of s_j' to s_j'' strictly raises the expected payoff, $u_j(\hat{\sigma}_j, \sigma_{-j}^*) > u_j(\sigma_j^*, \sigma_{-j}^*)$: a contradiction. Condition (2) follows in the same way. $\square$

Example 3: battle of the sexes in mixed strategies

		Chris: O	Chris: F
Alex	O	$2,\ 1$	$0,\ 0$
	F	$0,\ 0$	$1,\ 2$

Let Chris play O with probability $q \in [0,1]$. Alex's expected payoff from each action:

$$u_A(O,q) = 2q + 0\,(1-q) = 2q, \qquad u_A(F,q) = 0\,q + 1\,(1-q) = 1-q.$$

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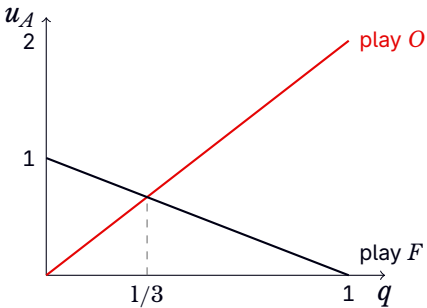
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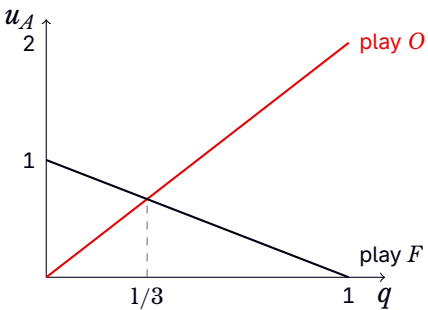
Alex prefers O if $2q > 1-q$, that is, if $q > \frac{1}{3}$.

Example 3: Alex's best response

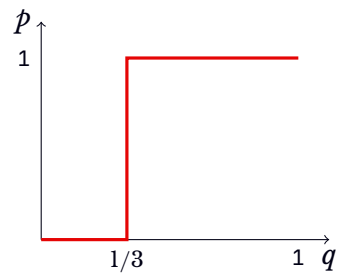


Expected payoff of each action of Alex

Example 3: Alex's best response



Expected payoff of each action of Alex



Alex's best response: probability p of O

$BR_A(q)$: F ($p = 0$) if $q < \frac{1}{3}$; any $p \in [0, 1]$ if $q = \frac{1}{3}$; O ($p = 1$) if $q > \frac{1}{3}$.

Example 3: Chris's best response

Let Alex play O with probability $p \in [0,1]$. Chris's expected payoff from each action:

$$u_C(O,p) = 1 \cdot p + 0\,(1-p) = p, \qquad u_C(F,p) = 0 \cdot p + 2\,(1-p) = 2 - 2p.$$

Example 3: Chris's best response

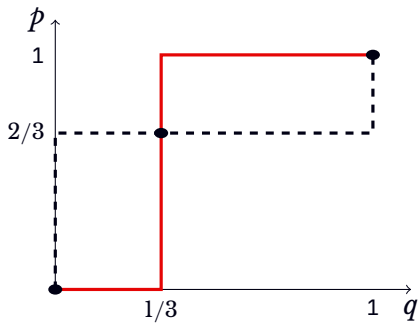
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Chris prefers O if $p > 2 - 2p$, that is, if $p > \frac{2}{3}$:

$$BR_C(p) = \begin{cases} q = 0 \ (F) & \text{if } p < \frac{2}{3}, \\ q \in [0,1] & \text{if } p = \frac{2}{3}, \\ q = 1 \ (O) & \text{if } p > \frac{2}{3}. \end{cases}$$

Example 3: all Nash equilibria

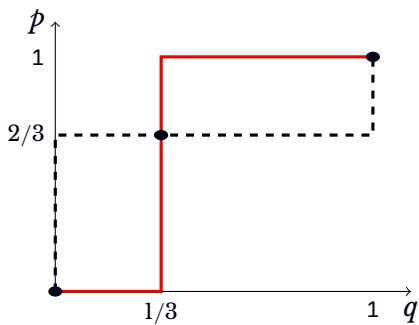


Best responses of Alex (solid) and Chris (dashed)

Equilibria = intersections of the best responses:

- $(p, q) = (0, 0)$: (F, F);
- $(p, q) = (1, 1)$: (O, O);
- $(p, q) = (\frac{2}{3}, \frac{1}{3})$: mixed equilibrium.

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In the mixed equilibrium, each probability is pinned down by the *other* player's payoffs; each player expects $2/3$, less than in either pure equilibrium.

Interpretations and existence

Common interpretations of mixed strategies:

- **Beliefs:** σ_{-i} describes what player i believes the others will do.
- **Empirical frequencies:** long-run frequencies of actions in repeated interactions.
- **Population distribution:** the share of a population of players using each action.

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Theorem (Nash, 1950)

Every finite normal-form game has a Nash equilibrium in mixed strategies.

Note: a profile of pure strategies is also a profile of mixed strategies (why?).

Application: monitoring remote work

Motivation: monitoring remote work

- In September 2025, Itaú dismissed employees after finding mismatches between their clock-in records and the activity monitored on their computers while working from home. The bank workers' union estimated about 1,000 dismissals.

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- Monitoring software is costly for firms, and employees know it may be in place.
- **Question:** how much should a firm monitor, and how much shirking should it expect?

Source: g1 (2025): Itaú demite funcionários após avaliar produtividade no home office

Mini case: work or shirk?

A remote employee chooses to **work** (W) or **shirk** (S); at the same time, the manager chooses to **monitor** (M) or **not** (N). A shirker who is caught loses the wage.

		Manager: M	Manager: N
Employee	W	$w-g,\ v-w-h$	$w-g,\ v-w$
	S	$0,\ -h$	$w,\ -w$

Parameters

$w = 10$	wage
$g = 2$	cost of effort
$v = 20$	value of output
$h = 1$	cost of monitoring

Mini case: questions (in pairs, hand in today)

(a) Show that the game has no Nash equilibrium in pure strategies.

(b) Find the mixed-strategy Nash equilibrium: the probability of shirking and the probability of monitoring.

(c) The firm buys cheaper monitoring software, and h falls from 1 to 0.5. What happens to shirking? And to monitoring? Explain the intuition.

(d) The firm raises the wage to $w = 20$. What happens to shirking and to monitoring?

(e) Write the equilibrium probabilities as functions of w, g and h. Which policy would you recommend to the firm, and which assumptions of the model would you question in the Itaú case?

Wrap-up and next lecture

Today

- Best responses, rationalizability and Nash equilibrium.
- Applications: coordination, prisoner's dilemma, Cournot and Bertrand.
- Mixed strategies, the principle of indifference and the existence of equilibrium; application to monitoring remote work.

Next lecture

- Dynamic games: extensive form, backward induction and subgame-perfect equilibrium.